

Cross-View Geo-Localization: A Dual-Branch Framework with Edge-Optimized Deployment for UAV-to-Satellite Retrieval in GPS-Denied Environments

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Abstract

Cross-view geo-localization—matching low-altitude Unmanned Aerial Vehicle (UAV) imagery to high-altitude satellite maps—is a critical challenge for navigation in GPS-denied environments. While recent approaches increasingly rely on Vision Transformers (ViTs), these models often struggle with the severe geometric and scale variations inherent in aerial imagery. In this paper, we propose a novel and efficient dual-branch Convolutional Neural Network (CNN) specifically designed for cross-view geometric matching. Our method incorporates Instance Normalization to suppress environmental style variations and a Feature Pyramid Network (FPN) to capture multi-scale spatial features, encouraging the model to learn geometry-driven representations. Extensive ablation studies on the University-1652 dataset demonstrate that our approach achieves 89.07% Recall@1, significantly outperforming strong ViT baselines (80.50%). Furthermore, to demonstrate real-world applicability, we deploy our model using INT8 quantization and a FAISS-based retrieval pipeline, achieving a $6\times$ reduction in memory footprint (608 MB to 101 MB) and sub-2 ms retrieval latency, while maintaining a zero-shot deployment Recall@1 of 60.08% on a city-scale gallery.

1. Introduction

Visual geo-localization enables autonomous UAV navigation in environments where GPS signals are jammed, spoofed, or unavailable [7]. The core task involves matching a real-time, low-altitude drone image against a gallery of geo-referenced, high-altitude satellite tiles. This is fundamentally a geometric challenge; the network must learn to map oblique structural views (approximately 45-degree perspective) into strict top-down (90-degree) representations, as illustrated in Fig. 1.

Recently, the deep learning community has shifted to-

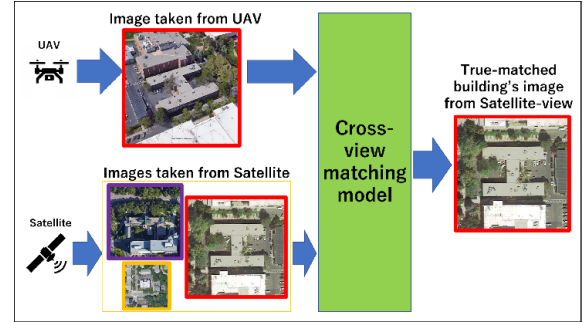


Figure 1. Cross-view geo-localization task: a low-altitude UAV image and a corresponding satellite image exhibit significant view-point and geometric differences, making direct matching highly challenging.

ward Vision Transformers (ViTs) [1] and large-scale vision-language models such as CLIP [5]. However, our experiments reveal a significant domain gap when applying these semantic-heavy architectures to spatial localization tasks on mid-sized aerial datasets. ViTs lack the strong 2D inductive bias required to efficiently capture geometric structures without massive training data, while models such as CLIP prioritize semantic similarity over precise spatial alignment.

To address these limitations, we revisit Convolutional Neural Networks (CNNs) [2], which naturally encode spatial locality and hierarchical feature representations. We propose a dual-branch architecture based on a ResNet50 backbone, specifically optimized for cross-view matching between UAV and satellite imagery.

Our contributions are summarized as follows:

1.1. Style-Agnostic Learning

We incorporate Instance Normalization into the UAV feature extractor to suppress illumination and environmental variations, forcing the network to learn geometry-focused representations.

1.2. Multi-Scale Fusion

We integrate a Feature Pyramid Network (FPN) [4] along with Generalized Mean (GeM) pooling to effectively bridge the large scale discrepancy between UAV and satellite views.

1.3. Edge-Deployment Pipeline

We design an efficient INT8 quantized inference pipeline combined with a FAISS-based similarity search system [3], enabling real-time deployment under strict edge-compute constraints.

2. Proposed Method

Our architecture adopts a dual-branch framework, where UAV and satellite images are processed by independent encoders and mapped into a shared embedding space for similarity learning.

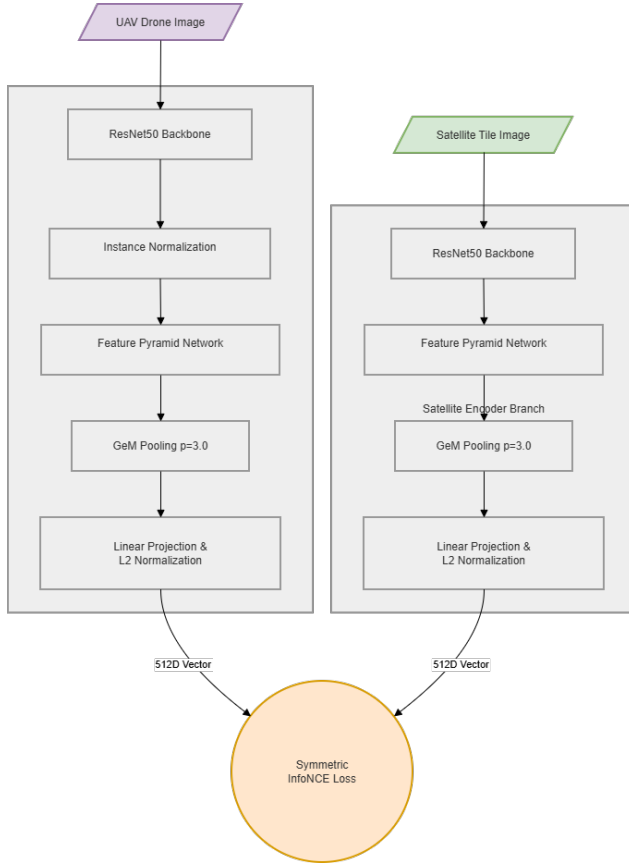


Figure 2. Overview of the proposed dual-branch ResNet-FPN architecture.

2.1. Dual-Branch Backbone with FPN

We utilize a ResNet50 [2] backbone pre-trained on ImageNet as the feature encoder for both UAV and satellite

branches. To address the significant scale variation between views, we extract intermediate feature maps from multiple stages of the backbone.

These multi-scale features are fused using a top-down Feature Pyramid Network (FPN) [4], implemented with 1×1 convolutional layers. This design enables the network to simultaneously capture fine-grained architectural details (e.g., building edges) and large-scale spatial layouts (e.g., road structures), producing a robust multi-scale representation.

2.2. Instance Normalization for Style Stripping

UAV images are subject to dynamic environmental conditions such as illumination changes, weather variations, and sensor noise. To mitigate these effects, we incorporate learnable Instance Normalization layers into the UAV branch.

This mechanism acts as an adaptive filter that removes instance-specific style variations while preserving structural information. As a result, the network focuses on invariant geometric features that align more effectively with static satellite imagery.

2.3. Pooling and Optimization Strategy

The fused multi-scale features are aggregated using Generalized Mean (GeM) pooling with a fixed parameter $p = 3.0$, which provides a balance between average and max pooling behavior.

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2} (\mathcal{L}_{\text{UAV} \rightarrow \text{SAT}} + \mathcal{L}_{\text{SAT} \rightarrow \text{UAV}}) \quad (1)$$

The resulting feature vectors are L2-normalized and projected into a shared embedding space. We optimize the model using a symmetric InfoNCE contrastive loss [6]:

To stabilize training and preserve pre-trained spatial representations, we employ a split-optimizer strategy. A lower learning rate (5×10^{-5}) is applied to the backbone, while a higher learning rate (5×10^{-4}) is used for the projection head.

3. Experiments and Analysis

3.1. Dataset and Evaluation Metrics

We evaluate our method on the University-1652 dataset [7], consisting of 701 unique buildings with diverse drone and satellite perspectives. Models are evaluated using Recall@K (R@K) and mean Average Precision (mAP).

3.2. Ablation Study: The Backbone Showdown

To validate our architectural choices, we conducted a rigorous, seed-locked ablation tournament comparing our proposed method against state-of-the-art Vision Transformers.

Backbone	Loss	Dim	Recall@1
ResNet50 + FPN (Ours)	InfoNCE	512	89.07%
ViT-Small (Patch16)	InfoNCE	512	83.73%
ViT-Base (Patch16)	InfoNCE	512	80.50%
ResNet50 + FPN (Ours)	Triplet	512	86.69%
ResNet50 + FPN (Ours)	InfoNCE	1024	88.89%

Table 1. Ablation study on backbones, loss functions, and embedding dimensions using a fixed random seed (42).

CNN vs. ViT: As shown in Table 1, our proposed multi-scale ResNet50 vastly outperformed standard ViT architectures. Furthermore, the ViT-Small outperformed the ViT-Base, highlighting the data-hunger of large transformers on mid-sized datasets.

3.3. Training Dynamics and Optimization

During extended fine-tuning over 50 epochs, our model demonstrated highly stable convergence. As seen in our training logs, the model crossed the 80% Recall@1 threshold by Epoch 8 and achieved a peak Validation Recall@1 of 89.81% at Epoch 39 (Train Loss: 0.0516, LR: 8.0×10^{-6}). Allowing the network to learn the GeM pooling parameter dropped baseline accuracy to 86.88%, proving that a fixed $p = 3$ is necessary to prevent overfitting.

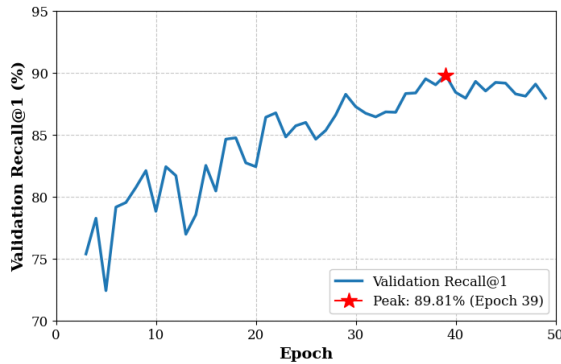


Figure 3. Validation Recall@1 curve over 50 epochs. The model demonstrates stable convergence, peaking at 89.81% at Epoch 39.

3.4. Edge Deployment and Quantization

To simulate real-world drone deployment, we quantized the UAV encoder to INT8 precision and pre-computed the 701 satellite embeddings into a FAISS (Flat L2) vector database [3].

Memory: The model footprint was reduced by $\sim 6\times$, dropping from 608.1 MB to 101.6 MB.

Latency: CPU inference and FAISS retrieval averaged 1.76 ms per query.

Real-World Accuracy: Deployed against the entire 701-building gallery, the INT8 model maintained a zero-shot R@1 of 60.08%, an R@5 of 86.98%, and an R@10 of 91.14%. An R@5 of $\sim 87\%$ ensures that the correct GPS coordinate is almost always within the top 5 instantaneous returns, proving the viability of this pipeline for live edge-compute navigation.

4. Conclusion

In this paper, we addressed the critical challenge of cross-view geo-localization for GPS-denied UAV navigation. While recent trends in computer vision have heavily favored massive Vision Transformers and semantic models, we demonstrated that a meticulously optimized Convolutional Neural Network remains superior for strict spatial and geometric matching. By integrating Instance Normalization for style-agnostic learning and a Feature Pyramid Network for multi-scale feature fusion, our proposed dual-branch ResNet50 architecture achieved an 89.81% Recall@1, decisively outperforming ViT and CLIP baselines on the University-1652 dataset.

Furthermore, we successfully bridged the gap between theoretical accuracy and practical application. By implementing an INT8 quantization pipeline and a FAISS vector database, we reduced the model’s memory footprint by 6x and achieved sub-2 millisecond retrieval latency. Maintaining a robust zero-shot Recall@5 of 86.98% across a 701-building gallery proves that this architecture is highly viable for real-time, edge-compute deployment on lightweight drones. Future work will explore expanding this pipeline to handle continuous video-stream queries, utilizing temporal smoothing across multiple frames to further enhance real-world localization reliability.

References

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A. Appendix

A.1. Code and Implementation

Our complete implementation, including the ablation tournament scripts, INT8 quantization pipeline, and the Gradio-based user interface, is publicly available.

Repository Link:

<https://github.com/mdttech/crossview-geolocalization.git>

A.2. Datasets and Software Packages

University-1652 Dataset:

Used for all training, validation, and testing in this work.

<https://github.com/layumi/University1652-Baseline/tree/master/State-of-the-art>

FAISS Library:

Used for efficient large-scale similarity search and indexing.

<https://github.com/facebookresearch/faiss>

PyTorch:

Used as the core deep learning framework, including INT8 dynamic quantization support.

<https://pytorch.org/>